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Open Geospatial Consortium

OGC MUDDI CONCEPTUAL MODEL

STANDARD

Conceptual model

DRAFT

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CONTENTS

I ABSTRACT

MUDDI (Model for Underground Data Definition and Interchange) is a representation of classes that serve as the basis that serve as the basis of integration of datasets that utilize different information models in the representation of underground objects.

MUDDI is meant to be comprehensive and provide sufficient level of detail to address application use cases, including but not limited to the following:

- routine street excavations;
- emergency response;
- utility maintenance programs;
- large scale construction projects;
- disaster planning;
- disaster response; and
- Smart Cities programs.

MUDDI serves as an excellent basis for conformant and interchangeable logical and physical implementations, such as GML (Geographic Markup Language) or SFS (Simple Features SQL).

This OGC Standard provides the full set of MUDDI conceptual models and their requirements.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, MUDDI, underground data

III PREFACE

This OGC Standard provides the MUDDI conceptual models as described in UML.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V SUBMITTERS

All questions regarding this document should be directed to the editor or the contributors:

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SCOPE

1 SCOPE

MUDDI (Model for Underground Data Definition and Interchange) is a representation of classes that serve as the basis that serve as the basis of integration of datasets that utilize different information models in the representation of underground objects.

MUDDI is intended to form the basis for conformant and interchangeable logical and physical implementations, such as GML (Geographic Markup Language) or SFS (Simple Features SQL).

This OGC Standard provides the full set of MUDDI conceptual models and their requirements.

2

CONFORMANCE

2 CONFORMANCE

This OGC Standard provides the MUDDI conceptual model.

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

There are no normative references in this document.

4

TERMS, DEFINITIONS AND ABBREVIATED TERMS

4 TERMS, DEFINITIONS AND ABBREVIATED TERMS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. Terms and definitions

4.1.1. **conceptual model**

CM ALTERNATIVE

model that defines concepts of a universe of discourse

[**SOURCE:** ISO 19101-1:2014, Clause 4.1.5]

4.1.2. **conceptual schema**

formal description of a *conceptual model* (Clause 4.1.1)

[**SOURCE:** ISO 19101-1:2014, Clause 4.1.6]

4.1.3. **model-driven standard**

MDS ALTERNATIVE

standard created using a model-driven architecture

[**SOURCE:** OGC 21-035r1, Clause 2.1.4]

4.1.4. conformance test case

test for a particular requirement or a set of related requirements

Note 1 to entry: When no ambiguity, the word “case” may be omitted. i.e. *conformance test* (Clause 4.1.7) is the same as *conformance test case* (Clause 4.1.4).

[**SOURCE:** OGC 08-131r3]

4.1.5. conformance test module

set of related tests, all within a single *conformance test class* (Clause 4.1.6)

[**SOURCE:** OGC 08-131r3]

4.1.6. conformance test class

conformance test level ALTERNATIVE

set of *conformance test modules* (Clause 4.1.5) that must be applied to receive a single certificate of conformance

[**SOURCE:** OGC 08-131r3]

4.1.7. conformance test

test, abstract or real, of one or more *requirements* (Clause 4.1.9) contained within a standard, or set of standards

[**SOURCE:** OGC 08-131r3]

4.1.8. recommendation

expression in the content of a document conveying that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others, or that a certain course of action is preferred but not necessarily required, or that (in the negative form) a certain possibility or course of action is deprecated but not prohibited

4.1.9. requirement

expression in the content of a document conveying criteria to be fulfilled if compliance with the document is to be claimed and from which no deviation is permitted

[**SOURCE:** OGC 08-131r3]

4.1.10. requirements class

aggregate of all *requirement modules* (Clause 4.1.11) that must all be satisfied to satisfy a *conformance test class* (Clause 4.1.6)

[**SOURCE:** OGC 08-131r3]

4.1.11. requirements module

aggregate of *requirements* (Clause 4.1.9) and *recommendations* (Clause 4.1.8) of a specification against a single *standardization target type* (Clause 4.1.15)

[**SOURCE:** OGC 08-131r3]

4.1.12. specification

document containing *recommendations* (Clause 4.1.8), *requirements* (Clause 4.1.9) and *conformance tests* (Clause 4.1.7) for those *requirements* (Clause 4.1.9)

4.1.13. standard

specification (Clause 4.1.12) that has been approved by a legitimate Standards Body

4.1.14. standardization target

entity to which some *requirements* (Clause 4.1.9) of a *standard* (Clause 4.1.13) apply

[**SOURCE:** OGC 08-131r3]

4.1.15. standardization target type

type of entity or set of entities to which the *requirements* (Clause 4.1.9) of a *standard* (Clause 4.1.13) apply

[**SOURCE:** OGC 08-131r3]

4.1.16. **statement**

expression in a document conveying information

[**SOURCE:** OGC 08-131r3]

4.2. Abbreviated terms

MDA	Model-Driven Architecture
MDS	Model-Driven Standard
OCL	Object Constraint Language
PIM	Platform Independent Model
PSM	Platform Specific Model
UML	Unified Modeling Language

5
CONVENTIONS

5 CONVENTIONS

5.1. General

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

5.2. Identifiers

The normative provisions in this standard are denoted by the URI

```
http://www.opengis.net/spec/{standard}/{m.n}
```

All requirements classes, requirements, conformance classes, and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

6

MODEL OVERVIEW

6 MODEL OVERVIEW

6.1. Design requirements

The development of MUDDI has been motivated by a number of specific design requirements:

Functionality	The model should specifically address three priority application use cases (Excavation, Large-scale Construction Design, Disaster Planning)
Compatibility	The model should derive from or map to/from existing candidate models such as INSPIRE [IUN] (Infrastructure for Spatial Information in the European Community), CityGML UN ADE [CUA] (Utility Networks Application Domain Extension), GeoSciML [GSML], or IMKL [IMKL] (Informatie Model Kabels en Leidingen).
Modularity	The model should have a modular structure with a simple mandatory core and additional elements in the form of optional extensions or interfaces, in order to minimize the quantity and detail of data required for specific use cases.
Traceability	The model should provide for UGII to be linked directly with the survey measurements, sensor observations, and supporting evidence from which it was derived and from which its quality can be determined.
Flexibility	The model should support implementations that include the geometric and topological representations needed for specific applications, such as 2D geometries for excavation planning, 3D geometries for construction design, or functional graph topologies for disaster vulnerability assessments.

6.2. Design patterns

There are a number of common model design patterns which might satisfy MUDDI requirements, particularly modularity and flexibility:

Relational	the pattern of fixed tables connected by foreign key relations. Clearly an established implementation pattern, but tends to result in duplication of data to support specialization and rather brittle relationships between model elements.

Graph	an extremely flexible pattern of data elements connected by diverse relation predicates tends to represent knowledge webs and physical networks very well, but is still hard to implement for spatial data.
Object	inheritance patterns are particularly well suited to UML modeling, but specialization hierarchies themselves can lack flexibility in terms of being able to choose different configurations of specialized attributes for specific applications.
Interface	a variant of the object model in which specializations can also be supported by the addition of interface elements (consisting of attributes and/or operations) to primary objects. Interfaces can be mixed and matched as well as reused for different primary objects, but the variety of ways in which they can be implemented may present a risk of non-interoperability between implementations.

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CONCEPTUAL MODEL REQUIREMENTS

7 CONCEPTUAL MODEL REQUIREMENTS

7.1. General

This clause specifies the requirements for the conceptual model.

7.2. Requirements of the Core conceptual model

The classes of the Core package of the conceptual model are:

- MUDDIObject
- MUDDIAsset
- MUDDIEvent
- MUDDISpace
- Network
- NetworkAsset
- NetworkConveyance
- NetworkNode
- NetworkLink
- NetworkAccessory

REQUIREMENTS CLASS 1: LOGICAL IMPLEMENTATION OF THE CORE CONCEPTUAL MODEL	
IDENTIFIER	`/req/core`
TARGET TYPE	Logical model
CONFORMANCE CLASS	Conformance class A.1: `/conf/core`

REQUIREMENTS CLASS 1: LOGICAL IMPLEMENTATION OF THE CORE CONCEPTUAL MODEL	
DESCRIPTION	A logical model implementation must implement the MUDDI Core conceptual model.
NORMATIVE STATEMENTS	Requirement 1: `/req/core/classes-concepts` Requirement 2: `/req/core/classes-associations` Requirement 3: `/req/core/classes-attributes` Requirement 4: `/req/core/concrete-value-types`
GUIDANCE	The MUDDI core conceptual models are illustrated in Figure 1.

REQUIREMENT 1: SPECIFICATION OF CONCEPTS IN THE CORE CONCEPTUAL MODELS	
IDENTIFIER	`/req/core/classes-concepts`
INCLUDED IN	Requirements class 1: `/req/core`
STATEMENT	For each UML class defined or referenced in the Core Package, the logical model SHALL contain an element which represents the same concept as that defined for the UML class.

REQUIREMENT 2: SPECIFICATION OF ASSOCIATIONS IN THE CORE CONCEPTUAL MODELS	
IDENTIFIER	`/req/core/classes-associations`
INCLUDED IN	Requirements class 1: `/req/core`
STATEMENT	For each UML class defined or referenced in the Core Package, the logical model SHALL represent associations with the same source, target, direction, roles, and multiplicities as those of the UML class.

REQUIREMENT 3: SPECIFICATION OF ATTRIBUTES IN THE CORE CONCEPTUAL MODELS	
IDENTIFIER	`/req/core/classes-attributes`
INCLUDED IN	Requirements class 1: `/req/core`
STATEMENT	For each UML class defined or referenced in the Core Package, the logical model SHALL represent the attributes of the UML class including the name, definition, type, and multiplicity.

REQUIREMENT 4: ABSTRACT VALUE TYPES IN CORE CONCEPTUAL MODELS REPLACED WITH CONCRETE VALUE TYPES	
IDENTIFIER	`/req/core/concrete-value-types`
INCLUDED IN	Requirements class 1: `/req/core`
STATEMENT	For each attribute specified with the AbstractValueType value type in a UML class defined or referenced in the Core Package, the logical model SHALL provide a concrete value type for that attribute that replaces the AbstractValueType value type.

7.3. Requirements for the extended conceptual models

The logical model may decide on which UML classes in the extended conceptual models to implement on an as needed basis.

In the Extended requirements class, there are no mandatory UML classes.

Therefore by default, if the minimum obligation in a relationship to a UML class is zero, then that UML class does not have to be implemented.

REQUIREMENTS CLASS 2: LOGICAL IMPLEMENTATION OF EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/req/extended`
TARGET TYPE	Logical model
CONFORMANCE CLASS	Conformance class A.2: `/conf/extended`
DESCRIPTION	A logical model implementation may implement any of the MUDDI extended conceptual models.
NORMATIVE STATEMENTS	Requirement 5: `/req/extended/classes-concepts` Requirement 6: `/req/extended/classes-associations` Requirement 7: `/req/extended/classes-attributes` Requirement 8: `/req/extended/concrete-value-types`
GUIDANCE	The MUDDI extended conceptual models are illustrated in Figure 2.

REQUIREMENT 5: SPECIFICATION OF CONCEPTS IN THE EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/req/extended/classes-concepts`
INCLUDED IN	Requirements class 2: `/req/extended`
STATEMENT	For each UML class adopted from the Extended Package to be implemented in the logical model, the logical model SHALL contain an element which represents the same concept as that defined for the UML class.

REQUIREMENT 6: SPECIFICATION OF ASSOCIATIONS IN THE EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/req/extended/classes-associations`
INCLUDED IN	Requirements class 2: `/req/extended`
STATEMENT	For each UML class adopted from the Extended Package to be implemented in the logical model, the logical model SHALL represent associations with the same source, target, direction, roles, and multiplicities as those of the UML class.

REQUIREMENT 7: SPECIFICATION OF ATTRIBUTES IN THE EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/req/extended/classes-attributes`
INCLUDED IN	Requirements class 2: `/req/extended`
STATEMENT	For each UML class adopted from the Extended Package to be implemented in the logical model SHALL represent the attributes of the UML class including the name, definition, type, and multiplicity.

REQUIREMENT 8: ABSTRACT VALUE TYPES IN THE EXTENDED CONCEPTUAL MODELS REPLACED WITH CONCRETE VALUE TYPES	
IDENTIFIER	`/req/extended/concrete-value-types`
INCLUDED IN	Requirements class 2: `/req/extended`
STATEMENT	For each attribute specified with the AbstractValueType value type in a UML class adopted from the Extended Package, the logical model SHALL provide a concrete value type for that attribute that replaces the AbstractValueType value type.

8

CONCEPTUAL MODEL PACKAGE

8 CONCEPTUAL MODEL PACKAGE

8.1. Conceptual Model overview

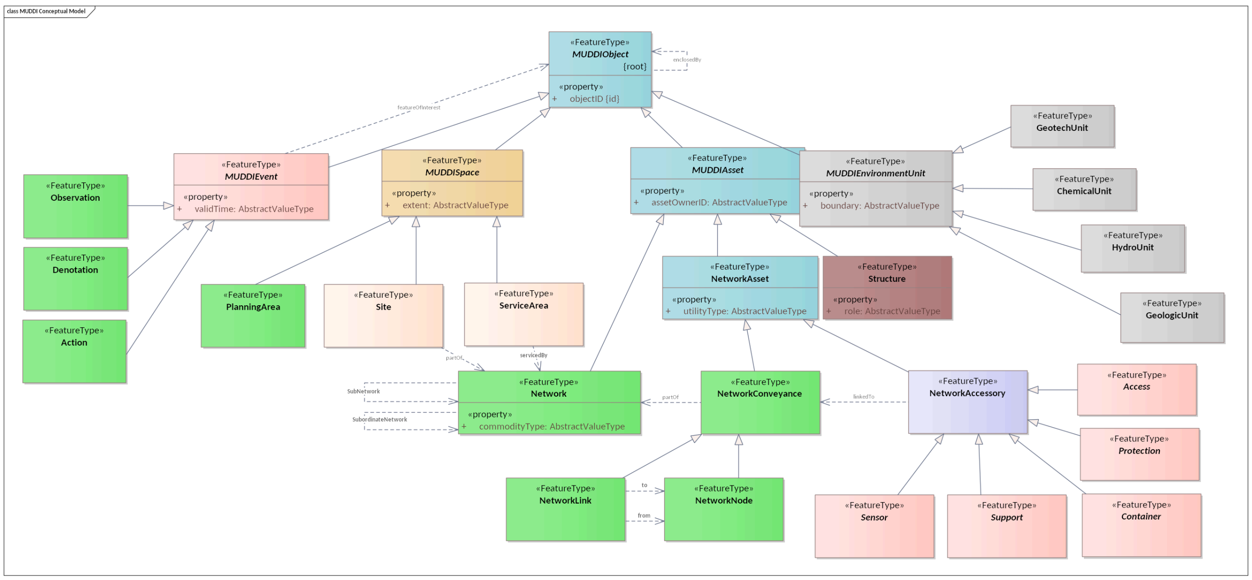

Figure 1 — MUDDI Conceptual Model

The key components of the MUDDI Conceptual Model are displayed in this diagram.

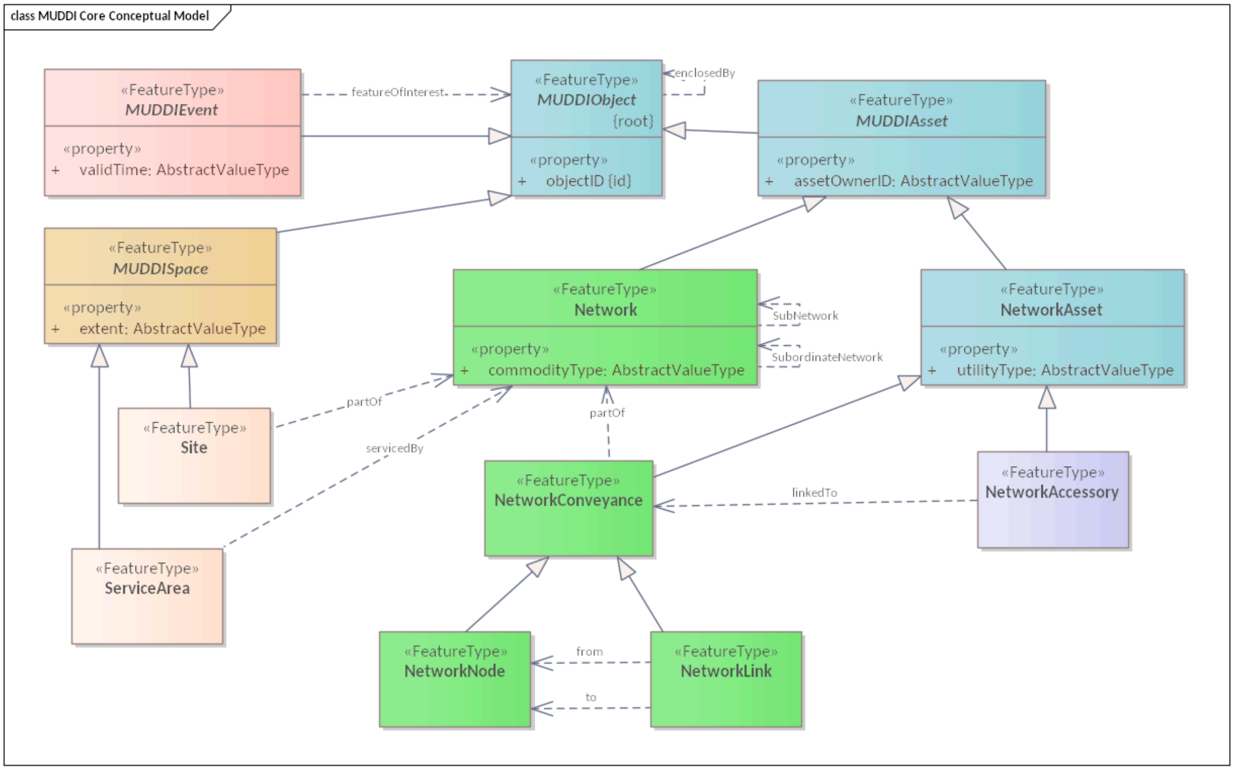

Figure 2 — MUDDI Core Conceptual Model

The components of the MUDDI Core Conceptual Model are displayed in this diagram.

8.2. Defining tables

Table 1 — Elements of “Conceptual Model::AbstractValueType” (Type)

NAME	AbstractValueType
DEFINITION	An abstract class used to represent an undetermined value type at the conceptual model level. Realization of attributes that accept values of this class should instead use a concrete value type.
STEREOTYPE	Type
ABSTRACT	True
ASSOCIATIONS	(none)
ATTRIBUTES	(none)

CONSTRAINTS	(none)

Table 2 — Elements of “Conceptual Model::Access” (FeatureType)

NAME	Access
DEFINITION	Type of NetworkAccessory that provides access to NetworkConveyances. This is generally an asset that connects the subsurface to the surface, such as a manhole.
STEREOTYPE	FeatureType
ABSTRACT	True
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 3 — Elements of “Conceptual Model::Action” (FeatureType)

NAME	Action
DEFINITION	Type of MUDDIEvent representing a task or other action such as a repair or calibration
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 4 — Elements of “Conceptual Model::ChemicalUnit” (FeatureType)

NAME	ChemicalUnit
DEFINITION	Environmental unit pertaining to chemical properties of subsurface materials
STEREOTYPE	FeatureType

ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 5 — Elements of “Conceptual Model::Container” (FeatureType)

NAME	Container
DEFINITION	Type of NetworkAccessory that contans one or more NetworkConveyances.
STEREOTYPE	FeatureType
ABSTRACT	True
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 6 — Elements of “Conceptual Model::Denotation” (FeatureType)

NAME	Denotation
DEFINITION	Type of MUDDIEvent representing the assigning of a particular value to a feature property, such as a status or sensitivity property.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 7 — Elements of “Conceptual Model::GeologicUnit” (FeatureType)

NAME	GeologicUnit
DEFINITION	Environmental unit pertaining to geological characterizations of subsurface materials
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 8 — Elements of “Conceptual Model::GeotechUnit” (FeatureType)

NAME	GeotechUnit
DEFINITION	Environmental unit pertaining to geotechnical properties of subsurface materials
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 9 — Elements of “Conceptual Model::HydroUnit” (FeatureType)

NAME	HydroUnit
DEFINITION	Environmental unit pertaining to hydrological and/or hydrogeological characteristics of subsurface materials
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)

ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 10 — Elements of "Conceptual Model::MUDDIAsset" (FeatureType)

NAME	MUDDIAsset		
DEFINITION	The parent class of built environment infrastructure that can be located, owned, operated, oriented, and consist of material.		
STEREOTYPE	FeatureType		
ABSTRACT	True		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	assetOwnerID [1]	AbstractValueType	Characteristic attribute of MUDDIAsset that identifies the present legal asset owner. Identifier maintenance and dereference mechanisms are implementation dependent.
CONSTRAINTS	(none)		

Table 11 — Elements of "Conceptual Model::MUDDIEnvironmentUnit" (FeatureType)

NAME	MUDDIEnvironmentUnit		
DEFINITION	The parent class of environmental units which represent regions of the subsurface exclusive of built assets		
STEREOTYPE	FeatureType		
ABSTRACT	True		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	boundary [1]	AbstractValueType	Characteristic attribute of MUDDIEnvironmentalUnit that represents the boundary in 2 or 3 dimensions delimiting each unit instance.

CONSTRAINTS	(none)

Table 12 — Elements of "Conceptual Model::MUDDIEvent" (FeatureType)

NAME	MUDDIEvent		
DEFINITION	The parent class of built environment events.		
STEREOTYPE	FeatureType		
ABSTRACT	True		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	validTime [1]	AbstractValue Type	Characteristic attribute of MUDDIEvent that represents the time interval during or instance at which the result of the event is valid. Usually but not always this is the same as or begins with the event occurrence.
CONSTRAINTS	(none)		

Table 13 — Elements of "Conceptual Model::MUDDIObject" (FeatureType)

NAME	MUDDIObject		
DEFINITION	The root class in MUDDI of underground infrastructure, space, event, environmental unit, or other underground +/- surface features that are identified, located, typed, and about which data is maintained.		
STEREOTYPE	FeatureType		
ABSTRACT	True		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	objectID [1]		Characteristic attribute of MUDDIObject that a persistent, unique identifier of the real world object represented by a MUDDIObject instance. Identifier maintenance and dereference mechanisms are implementation dependent but preference will be for global uniqueness.

CONSTRAINTS	(none)

Table 14 — Elements of "Conceptual Model::MUDDISpace" (FeatureType)

NAME	MUDDISpace		
DEFINITION	The parent class of defined, typically 2D regions comprising or overlying the subsurface.		
STEREOTYPE	FeatureType		
ABSTRACT	True		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	extent [1]	AbstractValueType	Characteristic attribute of MUDDISpace representing the typically 2D extent of each instance.
CONSTRAINTS	(none)		

Table 15 — Elements of "Conceptual Model::Network" (FeatureType)

NAME	Network		
DEFINITION	Type of MUDDIAsset representing the collection of network components comprising one network, which may be a subnetwork or subordinate network to another network.		
STEREOTYPE	FeatureType		
ABSTRACT	False		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	commodityType [1]	AbstractValueType	Characteristic attribute of Network that indicates the commodity being distributed by that Network instance.
CONSTRAINTS	(none)		

Table 16 — Elements of "Conceptual Model::NetworkAccessory" (FeatureType)

NAME	NetworkAccessory
DEFINITION	The parent class of network accessories that play a supporting role for NetworkConveyances.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 17 — Elements of "Conceptual Model::NetworkAsset" (FeatureType)

NAME	NetworkAsset		
DEFINITION	The parent class of network assets and components. \ Characteristic attribute*: utilityType		
STEREOTYPE	FeatureType		
ABSTRACT	False		
ASSOCIATIONS	(none)		
ATTRIBUTES	***NAME***	***TYPE***	***DEFINITION***
	utilityType [1]	AbstractValueType	Characteristic attribute of NetworkAsset that indicates the type of utility it forms a part of or commodity that it distributes.
CONSTRAINTS	(none)		

Table 18 — Elements of "Conceptual Model::NetworkConveyance" (FeatureType)

NAME	NetworkConveyance
DEFINITION	Type of MUDDIAsset that participates directly in distributing a commodity.
STEREOTYPE	FeatureType

ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 19 — Elements of "Conceptual Model::NetworkLink" (FeatureType)

NAME	NetworkLink
DEFINITION	Type of NetworkConveyance that forms a distribution channel between two NetworkNodes.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 20 — Elements of "Conceptual Model::NetworkNode" (FeatureType)

NAME	NetworkNode
DEFINITION	Type of NetworkConveyance that serves as a junction and connects two or more NetworkNodes to each other.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 21 — Elements of "Conceptual Model::Observation" (FeatureType)

NAME	Observation
DEFINITION	Type of MUDDIEvent representing an observation of phenomena leading to estimation of the value of an observableProperty of some FeatureOfInterest.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 22 — Elements of "Conceptual Model::PlanningArea" (FeatureType)

NAME	PlanningArea
DEFINITION	Represents the location and extent of region subject to specific policies, restrictions, and/or planning considerations.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 23 — Elements of "Conceptual Model::Protection" (FeatureType)

NAME	Protection
DEFINITION	Type of NetworkAccessory that provides protection such as cathodic protection to Network Conveyances.
STEREOTYPE	FeatureType
ABSTRACT	True

ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 24 — Elements of "Conceptual Model::Sensor" (FeatureType)

NAME	Sensor
DEFINITION	Type of NetworkAccessory comprising sensors or other ObservableProperty estimators.
STEREOTYPE	FeatureType
ABSTRACT	True
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 25 — Elements of "Conceptual Model::ServiceArea" (FeatureType)

NAME	ServiceArea
DEFINITION	Represents the typically 2D extent that is serviced by a particular network / subnetwork / subordinate network
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 26 — Elements of "Conceptual Model::Site" (FeatureType)

NAME	Site

DEFINITION	Represents the location and extent of a facility that is part of a network.
STEREOTYPE	FeatureType
ABSTRACT	False
ASSOCIATIONS	(none)
ATTRIBUTES	(none)
CONSTRAINTS	(none)

Table 27 — Elements of “Conceptual Model::Structure” (FeatureType)

<table>
<tr><td>NAME</td><td colspan="3">Structure</td></tr>
<tr><td>DEFINITION</td><td colspan="3">Type of built environment infrastructure that is not directly connected to a network but serves some structural role.</td></tr>
<tr><td>STEREOTYPE</td><td colspan="3">FeatureType</td></tr>
<tr><td>ABSTRACT</td><td colspan="3">False</td></tr>
<tr><td>ASSOCIATIONS</td><td colspan="3">(none)</td></tr>
<tr><td rowspan="2">ATTRIBUTES</td><td>NAME</td><td>TYPE</td><td>DEFINITION</td></tr>
<tr><td>role [1]</td><td>AbstractValueType</td><td>Characteristic attribute of Structure that indicates its role in the built environment such as basement.</td></tr>
<tr><td>CONSTRAINTS</td><td colspan="3">(none)</td></tr>
</table>

Table 28 — Elements of “Conceptual Model::Support” (FeatureType)

NAME	Support
DEFINITION	Type of NetworkAccessory that provides physical support to NetworkConveyances.
STEREOTYPE	FeatureType
ABSTRACT	True
ASSOCIATIONS	(none)

ATTRIBUTES	(none)
CONSTRAINTS	(none)

A

ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

A ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

A.1. General

This clause specifies the conformance classes and conformance tests for the conceptual model.

A.2. Conformance test suite of the Core conceptual model

CONFORMANCE CLASS A.1: TESTING THE LOGICAL IMPLEMENTATION OF THE CORE CONCEPTUAL MODEL	
IDENTIFIER	`/conf/core`
SUBJECT	Logical model
REQUIREMENTS CLASS	Requirements class 1: `/req/core`
DESCRIPTION	Test whether the logical model implementation fully implements the MUDDI Core conceptual model.
CONFORMANCE TESTS	Conformance test A.1: `/conf/core/classes-concepts` Conformance test A.2: `/conf/core/classes-associations` Conformance test A.3: `/conf/core/classes-attributes` Conformance test A.4: `/conf/core/concrete-value-types`

CONFORMANCE TEST A.1: TESTING LOGICAL MODEL CONCEPTS FROM CORE CONCEPTUAL MODELS	
IDENTIFIER	`/conf/core/classes-concepts`
REQUIREMENT	Requirement 1: `/req/core/classes-concepts`
INCLUDED IN	Conformance class A.1: `/conf/core`
TEST PURPOSE	To validate that the logical model correctly implements all UML classes defined or referenced in the Core Package.
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that the logical model contains a data element which represents the same concept as that defined for the UML class.

CONFORMANCE TEST A.2: TESTING LOGICAL MODEL ASSOCIATIONS FROM CORE CONCEPTUAL MODELS	
IDENTIFIER	`/conf/core/classes-associations`
REQUIREMENT	Requirement 2: `/req/core/classes-associations`
INCLUDED IN	Conformance class A.1: `/conf/core`
TEST PURPOSE	To validate that the logical model correctly implements all UML associations defined or referenced in the Core Package.
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that all UML associations in the Core package UML models are represented in the logical model. Validate that those relationships have the same source, target, direction, roles, and multiplicities as those documented in the Core package.

CONFORMANCE TEST A.3: TESTING LOGICAL MODEL ATTRIBUTES FROM CORE CONCEPTUAL MODELS	
IDENTIFIER	`/conf/core/classes-attributes`
REQUIREMENT	Requirement 3: `/req/core/classes-attributes`
INCLUDED IN	Conformance class A.1: `/conf/core`
TEST PURPOSE	To validate that the logical model correctly implements all UML attributes defined or referenced in the Core Package.

CONFORMANCE TEST A.3: TESTING LOGICAL MODEL ATTRIBUTES FROM CORE CONCEPTUAL MODELS	
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that all attributes of UML models from the Core package are represented in the logical model. Validate that those attributes have the same name, definition, type, and multiplicity as specified by the Core package.

CONFORMANCE TEST A.4: TESTING WHETHER ABSTRACT VALUE TYPES IN THE CORE CONCEPTUAL MODELS ARE REPLACED WITH CONCRETE VALUE TYPES	
IDENTIFIER	`/conf/core/concrete-value-types`
REQUIREMENT	Requirement 4: `/req/core/concrete-value-types`
INCLUDED IN	Conformance class A.1: `/conf/core`
TEST PURPOSE	To validate that the logical model correctly provides concrete value types for all UML attributes that are assigned the AbstractValueType value class.
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that all attributes of UML models from the Core package that utilize AbstractValueType are represented in the logical model with a concrete value type. Validate that there is no usage of the AbstractValueType class in the logical model.

A.3. Conformance test suite for the extended conceptual models

The logical model may decide on which UML classes in the extended conceptual models to implement on an as needed basis.

In the Extended requirements class, there are no mandatory UML classes.

Therefore by default, if the minimum obligation in a relationship to a UML class is zero, then that UML class does not have to be implemented.

CONFORMANCE CLASS A.2: TESTING THE LOGICAL IMPLEMENTATION OF EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/conf/extended`
SUBJECT	Logical model
REQUIREMENTS CLASS	Requirements class 2: `/req/extended`
DESCRIPTION	Test whether the logical model implementation of any of the MUDDI extended conceptual models conform to their specifications.
CONFORMANCE TESTS	Conformance test A.5: `/conf/extended/classes-concepts` Conformance test A.7: `/conf/extended/classes-associations` Conformance test A.6: `/conf/extended/classes-attributes` Conformance test A.8: `/conf/extended/concrete-value-types`

CONFORMANCE TEST A.5: TESTING LOGICAL MODEL CONCEPTS FROM THE EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/conf/extended/classes-concepts`
REQUIREMENT	Requirement 5: `/req/extended/classes-concepts`
INCLUDED IN	Conformance class A.2: `/conf/extended`
TEST PURPOSE	To validate that the logical model correctly implements the UML classes defined or referenced in the Extended Package.
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that for every UML class adopted from the Extended Package in the logical model, the logical model contains an element which represents the same concept as that defined for the UML class.

CONFORMANCE TEST A.6: TESTING LOGICAL MODEL ATTRIBUTES FROM THE EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/conf/extended/classes-attributes`
REQUIREMENT	Requirement 7: `/req/extended/classes-attributes`
INCLUDED IN	Conformance class A.2: `/conf/extended`
TEST PURPOSE	To validate that the logical model correctly implements all attributes from the adopted UML classes defined or referenced in the Extended Package.

CONFORMANCE TEST A.6: TESTING LOGICAL MODEL ATTRIBUTES FROM THE EXTENDED CONCEPTUAL MODELS	
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that all attributes of UML models adopted from the Extended package are implemented in the logical model. Validate that those attributes have the same name, definition, type, and multiplicity as specified by the Extended package.

CONFORMANCE TEST A.7: TESTING LOGICAL MODEL ASSOCIATIONS FROM THE EXTENDED CONCEPTUAL MODELS	
IDENTIFIER	`/conf/extended/classes-associations`
REQUIREMENT	Requirement 6: `/req/extended/classes-associations`
INCLUDED IN	Conformance class A.2: `/conf/extended`
TEST PURPOSE	To validate that the logical model correctly implements the adopted UML associations defined or referenced in the Extended Package.
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that all adopted UML associations from the Extended package are represented in the logical model. Validate that those relationships have the same source, target, direction, roles, and multiplicities as those documented in the Extended package.

CONFORMANCE TEST A.8: TESTING WHETHER ABSTRACT VALUE TYPES IN THE EXTENDED CONCEPTUAL MODELS ARE REPLACED WITH CONCRETE VALUE TYPES	
IDENTIFIER	`/conf/extended/concrete-value-types`
REQUIREMENT	Requirement 8: `/req/extended/concrete-value-types`
INCLUDED IN	Conformance class A.2: `/conf/extended`
TEST PURPOSE	To validate that the logical model correctly provides concrete value types for all UML attributes that are assigned the AbstractValueType value class.
TEST-METHOD-TYPE	Manual Inspection
DESCRIPTION	Validate that all attributes of UML models adopted from the Extended package that utilize AbstractValueType are represented in the logical model with a concrete value type. Validate that there is no usage of the AbstractValueType class in the logical model.

BIBLIOGRAPHY

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[1] Josh Lieberman, Open Geospatial Consortium: OGC 17-090r1, *Model for Underground Data Definition and Integration (MUDDI) Engineering Report*. Open Geospatial Consortium (2019). http://www.opengis.net/doc/PER/MUDDI-ER.

[2] Policy SWG, Open Geospatial Consortium: OGC 08-131r3, *The Specification Model — Standard for Modular specifications*. Open Geospatial Consortium (2009). https://docs.ogc.org/pol/08-131r3/08-131r3/pdf.

[3] Clemens Portele, Panagiotis (Peter) A. Vretanos, Charles Heazel, Open Geospatial Consortium: OGC 17-069r3, *OGC API — Features — Part 1: Core*. Open Geospatial Consortium (2019). http://www.opengis.net/doc/IS/ogcapi-features-1/1.0.0.

[4] Ronald Tse, Nick Nicholas, Open Geospatial Consortium: OGC 21-035r1, *OGC Testbed-17: Model-Driven Standards Engineering Report*. Open Geospatial Consortium (2022). http://www.opengis.net/doc/PER/t17-D022.

[5] Josh Lieberman, Andy Ryan, Open Geospatial Consortium: OGC 17-048, *OGC Underground Infrastructure Concept Study Engineering Report*. Open Geospatial Consortium (2017). http://www.opengis.net/doc/PER/uicds.

[6] International Organization for Standardization (committee): ISO 19101-1:2014, *Geographic information — Reference model — Part 1: Fundamentals*. International Organization for Standardization, Geneva (2014). https://www.iso.org/standard/59164.html.

[7] OMG UML 2.5, *Unified Modeling Language*. (2015). https://www.omg.org/spec/UML/2.5/About-UML.

[8] OMG XMI 2.5, *XML Metadata Interchange*. Object Management Group (2015). https://www.omg.org/spec/XMI/2.5.1/About-XMI/